

Proof versus Pattern Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	Time	30-40 min
Difficulty	Medium	ID	proof-versus-pattern

Big idea: Examples are not proof.

One-Minute Teacher Brief

Students distinguish empirical pattern spotting from mathematical proof.

Student Prompt

Inspect Proof versus Pattern Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Proof versus Pattern Stimulus A	Student-facing stimulus 1: A number pattern such as $1 + 2 + \dots + n = n(n+1)/2$ that invites both pattern-spotting and proof. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Proof versus Pattern Stimulus B	Student-facing stimulus 2: A visual pattern students extend before testing whether it proves anything. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Proof versus Pattern Reveal / Twist	Student-facing stimulus 3: A comparison between inductive confidence and deductive proof. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Examples are not proof.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Proof versus Pattern Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Proof versus Pattern Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Why is proof valued more than observation in mathematics?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how examples are not proof.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Why is proof valued more than observation in mathematics?

Setup Checklist

- Prepare a starter stimulus using: A number pattern such as $1 + 2 + \dots + n = n(n+1)/2$ that invites both pattern-spotting and proof.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- A number pattern such as $1 + 2 + \dots + n = n(n+1)/2$ that invites both pattern-spotting and proof.
- A visual pattern students extend before testing whether it proves anything.
- A comparison between inductive confidence and deductive proof.

Resources To Use

- Resource pack: <resources/proof-versus-pattern/index.html>
- Card deck CSV: <resources/proof-versus-pattern/cards.csv>
- Visual prompt: <resources/proof-versus-pattern/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Students calculate $n^2 + n + 41$ for $n = 0, 1, 2, \dots$ and record whether each output is prime.
2. They predict whether the expression always gives primes.
3. Reveal a counterexample, such as $n = 40$.
4. Students write the difference between many examples and proof.

Debrief Moves

- Knowledge question: Why is proof valued more than observation in mathematics?
- Can something be obvious but still require proof?
- Is mathematics discovered or invented?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...